

RISHAB MANTRI

Senior Software Engineer | Distributed Systems · GenAI & Agentic AI · System Design

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PROFESSIONAL SUMMARY

Senior software engineer with 6+ years architecting and scaling distributed systems at Amazon. Owns core data infrastructure for a \$7.2B-GMS platform spanning 900+ brands, has shipped \$130K+ in annualized cost savings and an 80%+ reliability turnaround, and builds production GenAI and agentic AI systems on AWS Bedrock, SageMaker, and MCP. Repeatedly tapped to step into ambiguous, cross-team ownership gaps and deliver ahead of deadline.

PROFESSIONAL EXPERIENCE

Software Engineer II — Amazon (Selling Partner Experience) Dec 2022 – Present

- Re-architected the core data layer powering a **\$7.2B-GMS platform across 900+ brands**, improving scalability and cross-team API consistency; the design was adopted as the org's scaling standard.
- Led Vine Pre-Release across 6 cross-functional teams to cut time-to-first-review **72% (39 to 11 days)**, ship **51%** of enrolled products with day-one reviews, and launch 2 months ahead of the Amazon Accelerate deadline.
- Eliminated **\$90K/month** in seller losses by designing cancellation-prevention logic that blocked **1,500 non-compliant cancellations/month**, then cut CSA ticket volume **80%** (60 to 10 per week) with a purpose-built investigation tool.
- Architected and led a 4-engineer team building an automated trust & safety pipeline (Step Functions, Lambda) that replaced a manual quarterly process, flagging **200 policy-violating reviewers** in its first run via a 3-strike enforcement model.
- Built and scaled a GenAI feedback platform on **AWS Bedrock + Claude** from a solo hackathon prototype into org-wide infrastructure, eliminating **~2 hrs/week** of manual review per person across ~500 weekly entries.
- Designed an explainable ML pipeline (**SageMaker**, Apache Airflow) diagnosing seller underperformance across 15 root-cause categories with automated retraining, lifting recommendation acceptance **200%+**.
- Integrated agentic AI into Seller Assistant via **MCP (Model Context Protocol)**, letting sellers query ASIN benchmarking data conversationally; launched as a full-scale beta to **all U.S. sellers**.
- Mentored engineers and led hands-on **AI-fluency workshops** across the team & org, upskilling colleagues on applied GenAI and agentic AI techniques to keep the org current with fast-moving AI trends.
- Directed international expansion into India, Brazil, and Mexico as sole technical advisor, onboarding 8 engineers across 3 markets and driving **25K enrollments** and **40K orders** within the first 2 months of launch.
- Drove **\$80K+** in annualized infrastructure savings as team cost champion, right-sizing CloudWatch logging and retention across an 8-engineer team.

Software Engineer I — Amazon Aug 2020 – Nov 2022

- Overhauled the CXM Elasticsearch pipeline (Lambda, S3, ES) after data scaled from 370GB to 1TB; self-identified the bottleneck and cut sync time **67% (9 to 3 hours)** and throttling fatals from 272/month to **zero**, earning promotion to Software Engineer II.
- Deprecated the underused Brand Catalog Manager and led a 4-engineer team to rebuild its value as a streamlined tab, saving **\$50K/year** and ~1 headcount while preserving functionality for **1,200+ managed brands**.
- Owned the image-validation component for a cross-team multi-ASIN email campaign, driving a **286%** lift in weekly OPS and **91.3%** growth in email sends.

TECHNICAL SKILLS

Languages: Java (Advanced), Python, JavaScript

Cloud & Infrastructure: AWS (Lambda, S3, Bedrock, SageMaker, Step Functions, SQS, DynamoDB, Elasticsearch, CloudWatch, IAM), Apache Airflow

AI / ML: GenAI, Agentic AI, MCP (Model Context Protocol), AWS Bedrock, Claude, LLM Pipelines, Model Retraining & Explainability

Frameworks & Tools: Spring Boot, Node.js, React, .NET, REST APIs, Git, Jira

Specialties: Distributed Systems, Microservices, System Design, Cross-Team Technical Leadership, Trust & Safety Automation, Cost Optimization

EDUCATION

M.S., Software Engineering — Arizona State University, Tempe, AZ 2019

B.E., Computer Science — Shri Ramdeobaba College of Engineering, Nagpur, India 2017